

Gluing Equations in Low Dimensional Geometry

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(joint with A. Casella)

Oklahoma State Topology Seminar

Nov 4, 2020

Overview

1. Geometric Structures

- What are they?
- Why are they interesting?
- How do we construct them?

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- Tool for constructing geometric structures
(*or representations*)
- Philosophy: Gluing equations “discretize” the problem of constructing geometric structures

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- Tool for constructing geometric structures
(*or representations*)
- Philosophy: Gluing equations “discretize” the problem of constructing geometric structures
- Examples
 - Thurston’s equations
 - *Garoufalidis-Goerner-Zickert $\mathrm{PGL}_n(\mathbb{C})$ gluing equations*
 - *Bergeron-Falbel-Guilloux $\mathrm{PGL}_3(\mathbb{C})$ gluing equations*
 - B-Casella projective gluing equations

What is geometry?

Super accurate historical reenactment

Felix Klein
Erlangen, Germany
(circa 1872)

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Geometry is the study of properties of X that are invariant under G

Some Examples

Metric geometries

1. $(\mathbb{E}^n, \text{Isom}(\mathbb{E}^n))$ (Euclidean geometry)
2. $(\mathbb{H}^n, \text{Isom}(\mathbb{H}^n))$ (hyperbolic geometry)
3. $(\mathbb{S}^n, \text{Isom}(\mathbb{S}^n))$ (spherical geometry)

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Geometric Properties: distance, angles, volumes, geodesics

Some Examples

Affine geometries

1. $(\mathbb{R}^n, \mathbb{R}^n \rtimes \mathrm{GL}_n(\mathbb{R}))$ (affine geometry)
2. $(\mathbb{R}^n, \mathbb{R}^n \rtimes \mathrm{SL}_n(\mathbb{R}))$ (special affine geometry)
3. $(\mathbb{R}^n, \mathbb{R}^n \rtimes O(n))$ (Euclidean geometry)
4. $(\mathbb{R}^n, \mathbb{R}^n \rtimes O(n-1, 1))$ (Lorentz Geometry)

$$\mathbb{R}^n \rtimes \mathrm{GL}_n(\mathbb{R}) \curvearrowright \mathbb{R}^n \text{ via } (b, A) \cdot x = Ax + b$$

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Subgeometries: 2-4 are *subgeometries* of 1. (there are compatible embeddings of the spaces and groups)

Some Examples

Projective geometry

- $\mathbb{RP}^n$ is the space of lines through the origin in $\mathbb{R}^{n+1}$
- $\mathrm{PGL}_{n+1}(\mathbb{R}) = \mathrm{GL}_{n+1}(\mathbb{R})/\text{scaling}$
- *(Real) projective geometry* is $(\mathbb{RP}^n, \mathrm{PGL}_{n+1}(\mathbb{R}))$ geometry

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Projective geometry contains lots of interesting subgeometries!

Projective geometry

Subgeometries

Affine geometries are subgeometries

- $\mathbb{R}^n \hookrightarrow \mathbb{RP}^n := \mathbb{R}^n \sqcup \mathbb{RP}^{n-1},$
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Gives unified setting to study different geometries

(X, G) structures

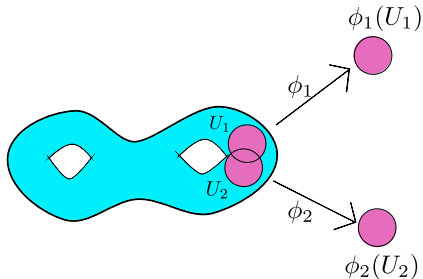
Definition

An (X, G) *structure* on a manifold M is a (maximal) atlas of charts from M into X whose transition functions are elements of G

(X, G) structures

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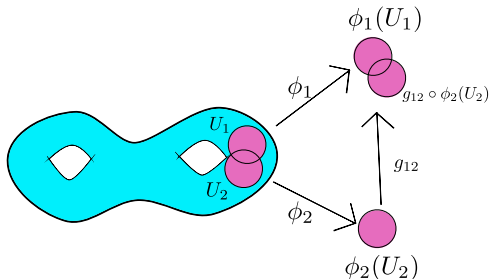
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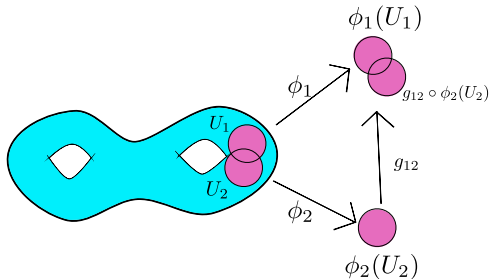
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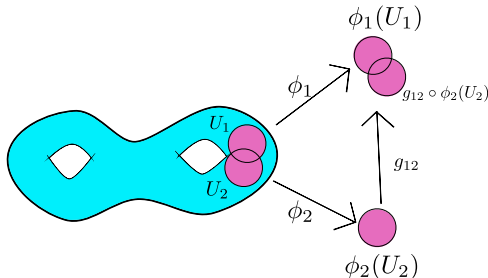


For $z \in U_1 \cap U_2$, $\phi_1(z) = g_{12}\phi_2(z)$

(X, G) structures

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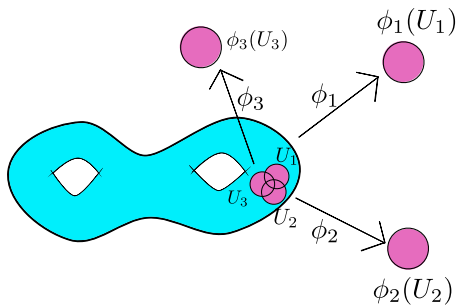
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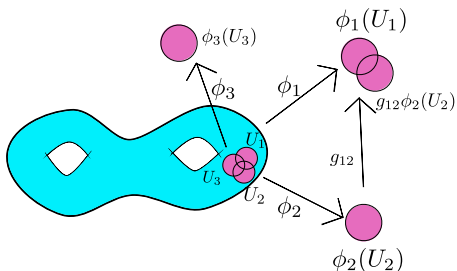
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Analyticity ensures that transition functions are **unique!**

Cocycle condition

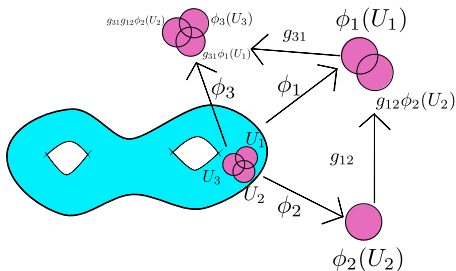


Cocycle condition



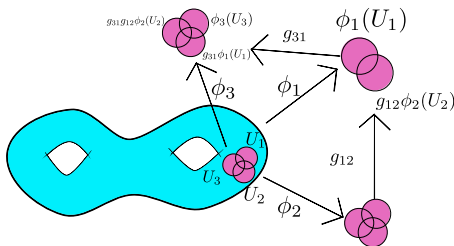
$$\text{For } z \in U_1 \cap U_2 \cap U_3, \quad g_{12}\phi_2(z) = \phi_1(z) \quad (1)$$

Cocycle condition



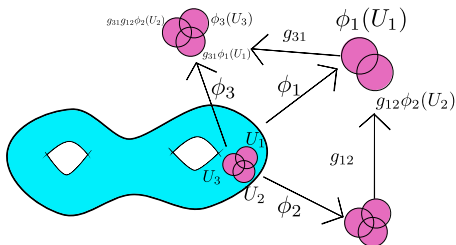
$$\text{For } z \in U_1 \cap U_2 \cap U_3, \quad g_{12}\phi_2(z) = \phi_1(z) \Rightarrow \\ g_{31}g_{12}\phi_2(z) = g_{31}\phi_1(z) = \phi_3(z) \quad (1)$$

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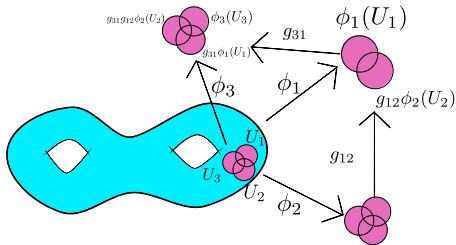
$$\begin{aligned}
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Cocycle condition



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 $g_{23}g_{31}g_{12} = 1$ **Cocycle condition**

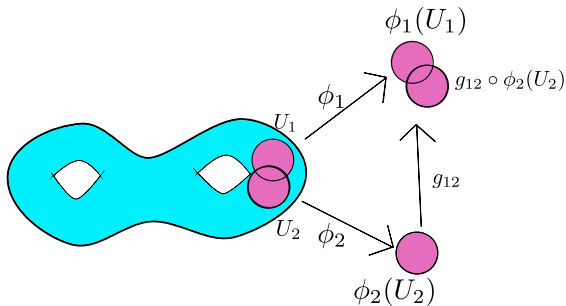
Cocycle condition



Conversely, a covering by open sets and a collection of maps satisfying this cocycle condition are the transition maps for an (X, G) structure!

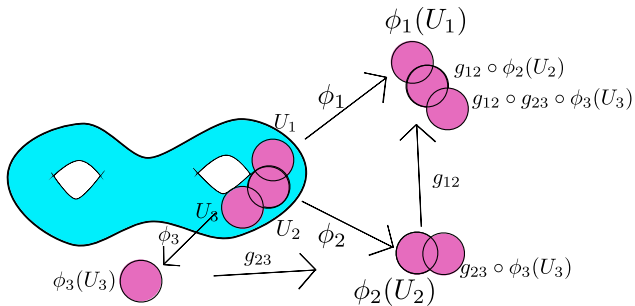
Development and holonomy

A more global approach



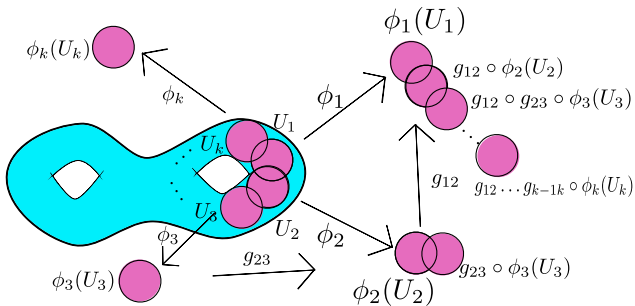
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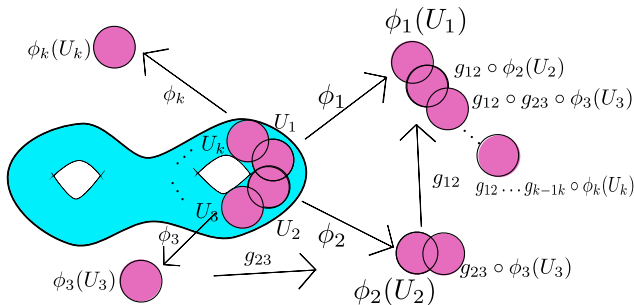
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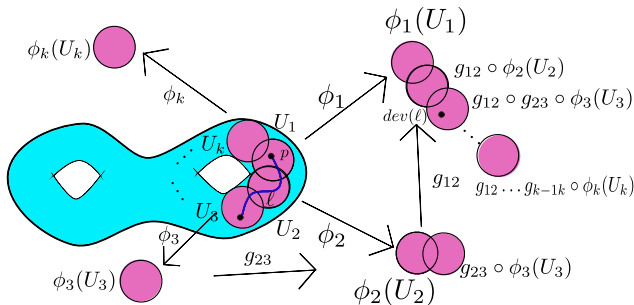
Not well defined on M , We are really defining

$$\text{dev} : \tilde{M} \rightarrow \mathbb{RP}^n,$$

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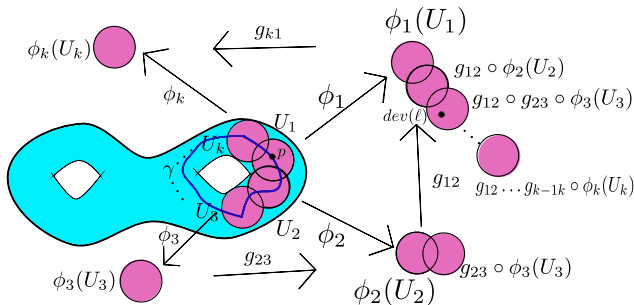
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$$[\gamma] \mapsto (g_{12} \dots g_{k1})^{-1}$$

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Properties

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- dev is a **hol-equivariant local diffeomorphism**
i.e. $\text{dev}(\gamma \cdot x) = \text{hol}(\gamma) \cdot \text{dev}(x) \quad \forall x \in \tilde{M}, \gamma \in \pi_1 M$

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- Constructing a projective structure is equivalent to constructing such an equivariant pair

Gluing equations

Rough idea

Problem: The space of all possible charts is huge (*infinite dimensional*), lots of redundancy

Gluing equations

Rough idea

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Solution: Use a triangulation to reduce to a manageable set of charts (*finite dimensional*) and use finitely many equations to check *cocycle condition*

Gluing equations

Ideal triangulations

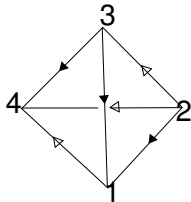
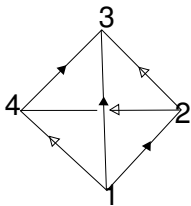
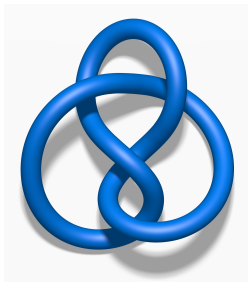
Let M be a non-compact 3-manifold with a finite ideal (no vertices) triangulation $\mathcal{T}$

Gluing equations

Ideal triangulations

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Ex: Figure-8 knot complement



Gluing equations

Variables and equations

Idea: Restrict charts so that their domains are tetrahedra in $\mathcal{T}$ and the maps are simplicial maps to “straight” tetrahedra in $\mathbb{RP}^3$

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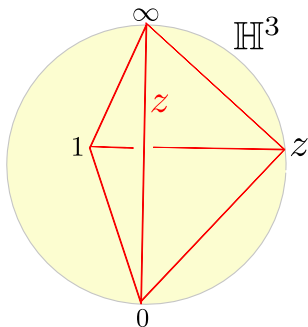
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Equations

- Constraints must be imposed on shape parameters to ensure compatibility of charts + cocycle condition
 - i. **Face equations:** Ensure two tetrahedra can be glued together along a face (**existence of transition maps**)
 - ii. **Edge equations:** Ensure that the tetrahedra abutting an edge in $\mathcal{T}$ close up in $\mathbb{RP}^3$ (**cocycle condition**)

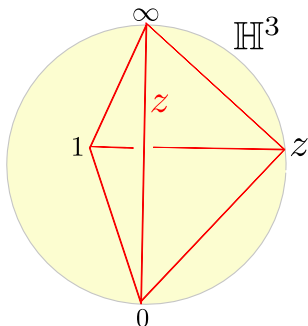
Thurston's gluing equations

- Klein model gives $\mathbb{H}^3 \subset \mathbb{RP}^3$ and identifies $\partial\mathbb{H}^3$ with $\mathbb{C} \cup \infty$
- **Fact:** $\text{Isom}^+(\mathbb{H}^3)$ acts simply triply transitively on $\partial\mathbb{H}^3$



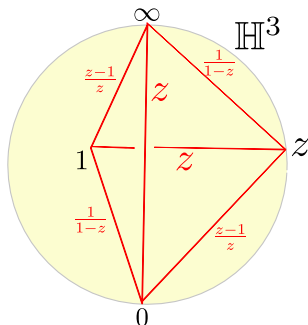
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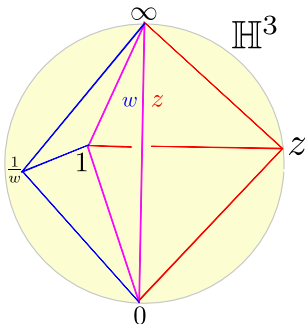
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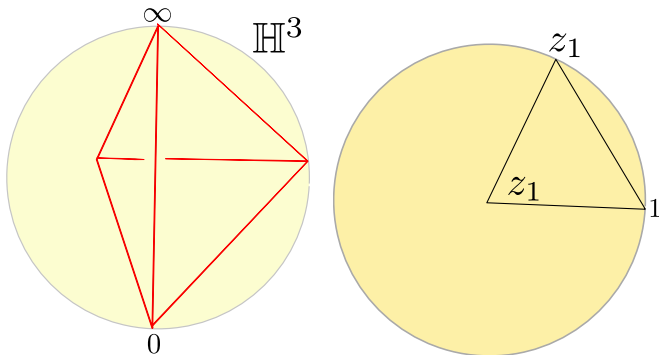
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 - Given two ideal tetrahedra, there is a unique way to glue them along any face



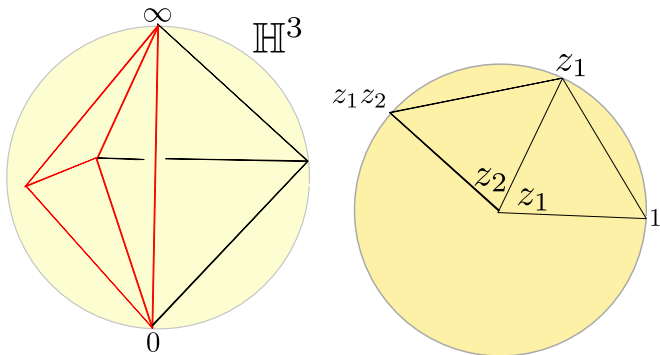
Thurston's gluing equations

Given a collection of ideal tetrahedra, we can glue them together around an edge



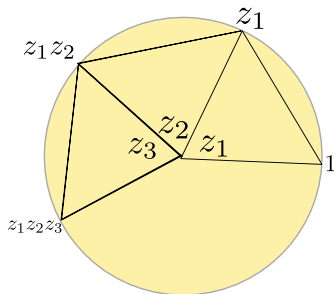
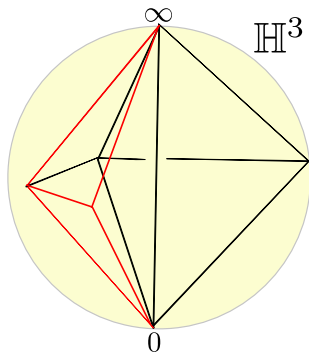
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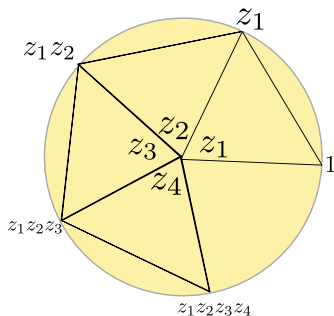
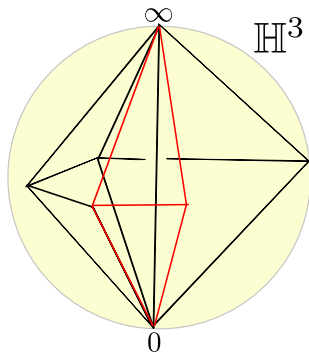
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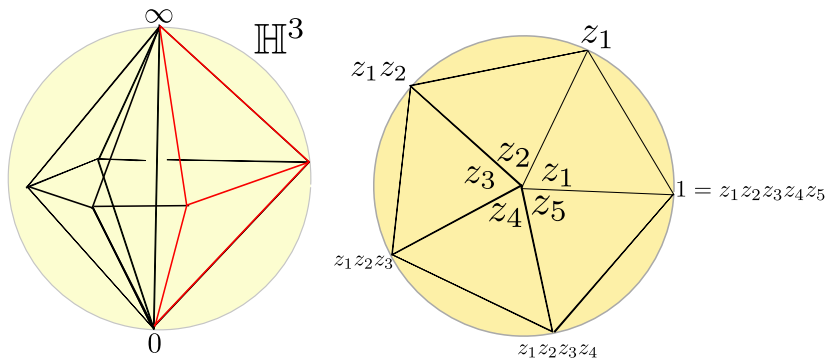
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In order for the cycle to close up we need to impose an equation (*cocycle condition*)

Thurston's gluing equations

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(*Thurston's gluing equations*)

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- Variables:
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- Equations:
 - no face equations
 - 1 edge equation for each edge in $\mathcal{T}$.

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(*Thurston's gluing equations*)

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A solution to these equations is *geometric* if each component has positive imaginary part (**No inside out tetrahedra**)

Naive approach for projective structures

Let's just try to build tetrahedra in $\mathbb{RP}^3$ like we did for $\mathbb{H}^3$.

Naive approach for projective structures

Let's just try to build tetrahedra in $\mathbb{RP}^3$ like we did for $\mathbb{H}^3$.

Good news

- All tetrahedra are projectively equivalent
(No shape parameters)
- Any two tetrahedra can be glued along a face
(no face equations)

Naive approach for projective structures

Let's just try to build tetrahedra in $\mathbb{RP}^3$ like we did for $\mathbb{H}^3$.

Good news

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(No shape parameters)
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Bad news

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- Tetrahedra in $\mathbb{RP}^3$ are not uniquely determined by their vertices

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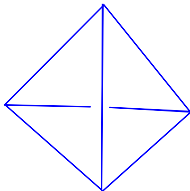
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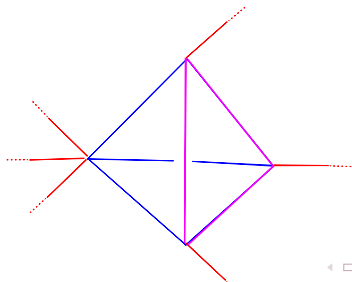
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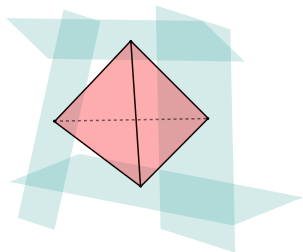
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Tetrahedra of flags

A *tetrahedron of (incomplete) flags in $\mathbb{RP}^3$* consists of

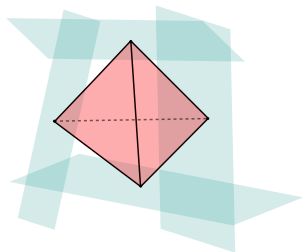
- 4 points in $\mathbb{RP}^3$, (v_1, v_2, v_3, v_4)
- A plane through each point, (P_1, P_2, P_3, P_4)
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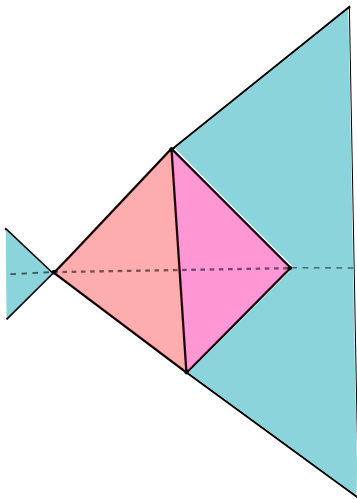


Don't have the previous problems for tet of flags

- Tetrahedron of flags has trivial PGL_4 stabilizer.
- A tetrahedron of flags determines a unique tetrahedron in $\mathbb{RP}^3$

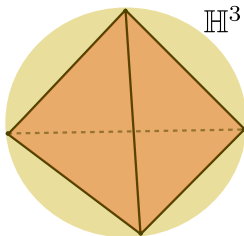
Why tetrahedra of flags?

Tetrahedra in $\mathbb{RP}^3$ are not determined by vertices



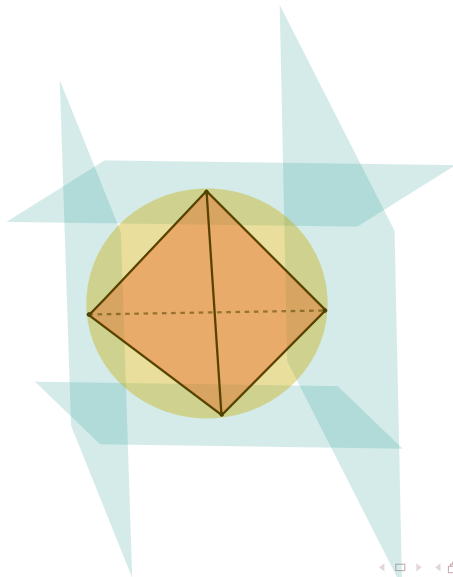
Why tetrahedra of flags?

$\mathbb{H}^3$ allows us to single out one of these tetrahedra



Why tetrahedra of flags?

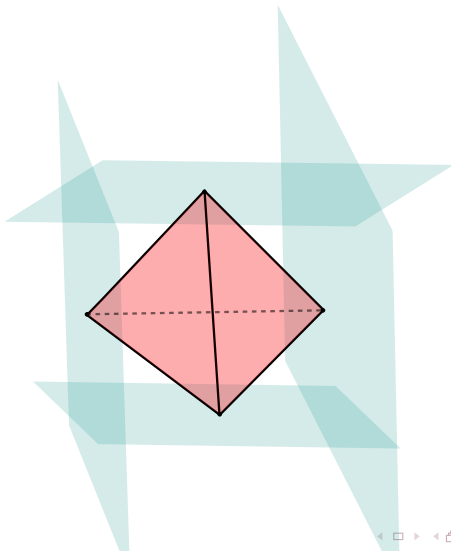
An ideal tet in $\mathbb{H}^3$ determines a tet of flags via tangent planes



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An ideal tet in $\mathbb{H}^3$ determines a tet of flags via tangent planes

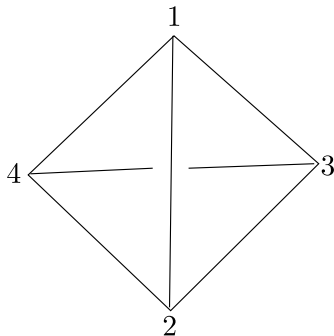
A tet of flags doesn't require planes to be tangent to $\mathbb{H}^3$



Projective gluing equations

Variables

Each tetrahedron of flags comes with coordinates

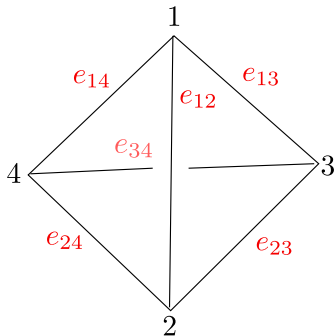


Projective gluing equations

Variables

Each tetrahedron of flags comes with coordinates

- **6 Edge coordinates:** 1 per edge: Describe the shape of the tetrahedron of flags



Let (ijk) be even perm of (1234) the

$$e_{ij} = e_{ji} = \frac{P_i(V_k)P_j(V_l)}{P_i(V_l)P_j(V_k)}$$

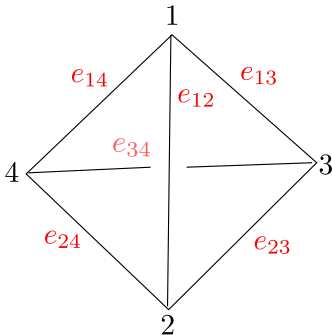
tet of flags $\Leftrightarrow$ all coordinates are positive real numbers

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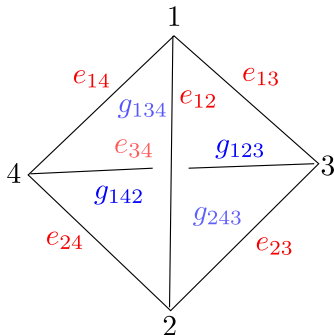
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Projective gluing equations

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- 4 Gluing coordinates: 1 per face: Describe how adjacent tetrahedra of flags will be attached.



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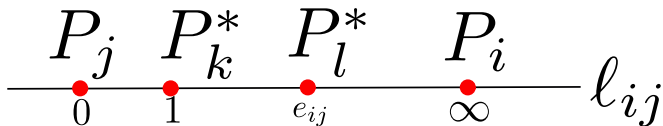
Geometric interpretation of edge coordinates

- ℓ_{ij} = pencil of planes containing $P_i \cap P_j$
- P_k^* = plane in ℓ_{ij} containing V_k
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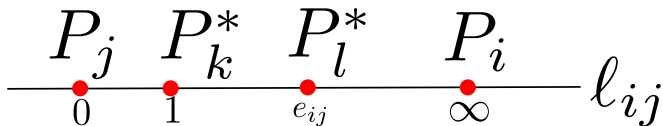


$$e_{ij} = [P_i, P_j, P_k^*, P_l^*]$$

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$$e_{ij} = [P_i, P_j, P_k^*, P_l^*]$$

Moral: If you know e_{ij} and you know where P_i , P_j , and P_k^* are you can locate P_l^*

Projective gluing equations

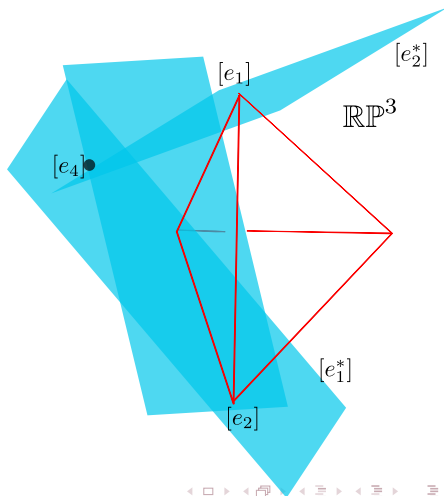
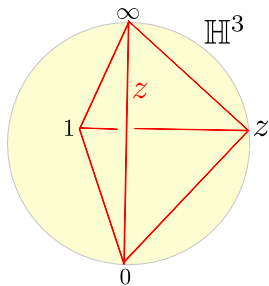
Standard position

Similar to ideal hyperbolic tets there is a standard position for tets of flags

Projective gluing equations

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Projective gluing equations

Face equations

There are two types of face equations

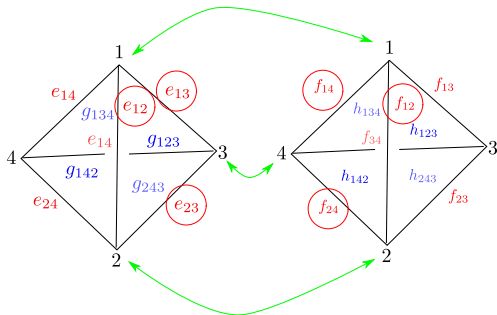
Projective gluing equations

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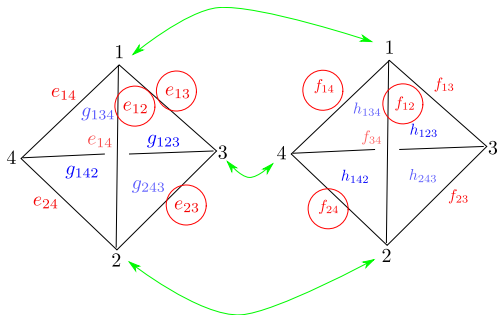
Projective gluing equations

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- $\prod \text{red} = 1$
- ensures the triples of flags are projectively equivalent (*existence of transition functions*)

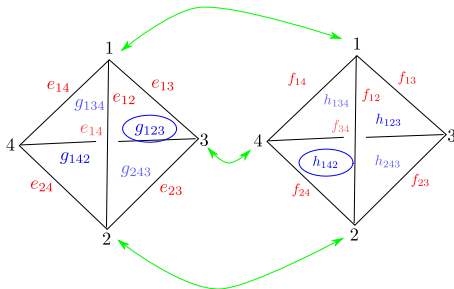


Projective gluing equations

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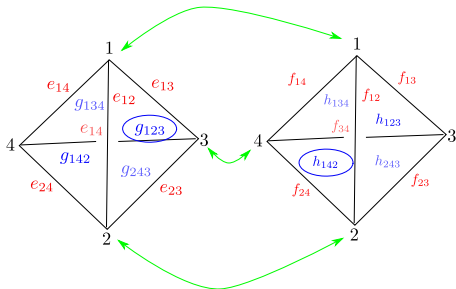
Projective gluing equations

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Type 2

- gluings between triples of flags are not unique
(1 *parameter family*)



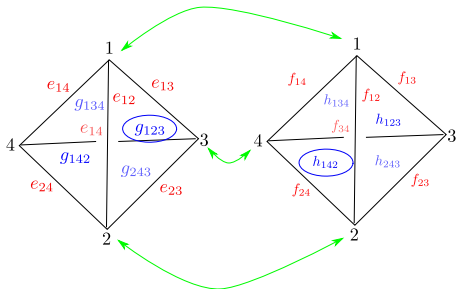
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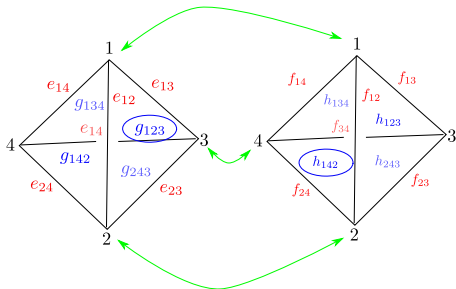
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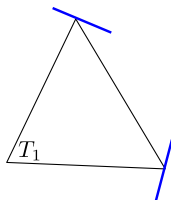
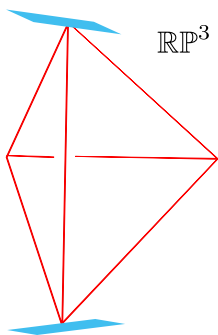
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Projective gluing equations

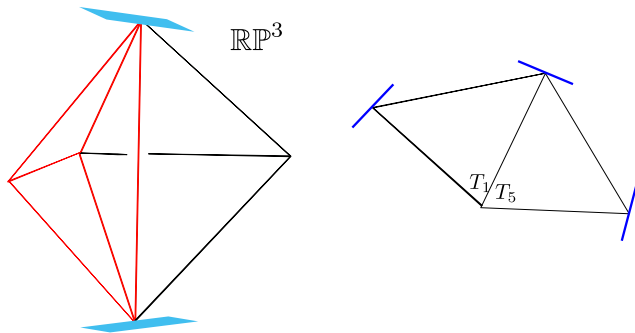
Edge equations



Let $T_1, \dots, T_n$ be a cycle of tets of flags abutting an edge in $\mathcal{T}$
Let $G_i \in \mathrm{PGL}_4$ moves T_{i+1} in std pos and glue to T_i in std pos

Projective gluing equations

Edge equations

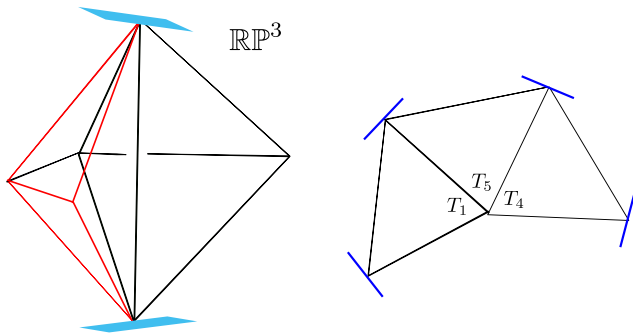


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$$G_5(T_1)$$

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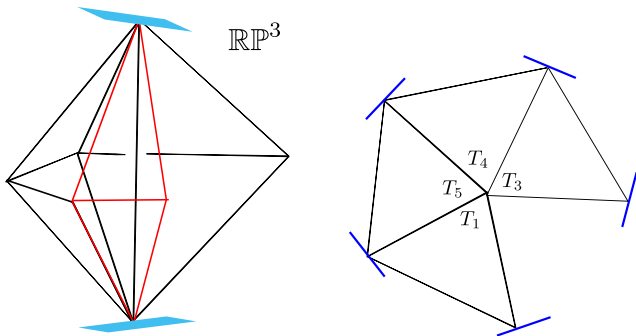


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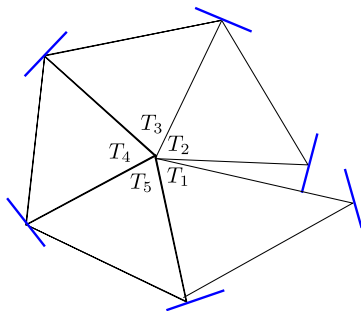
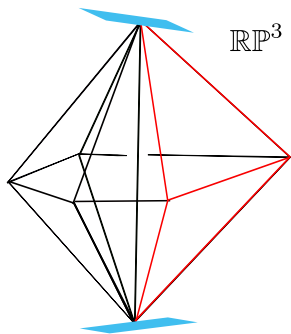


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Projective gluing equations

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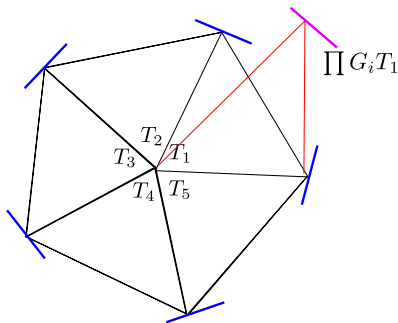
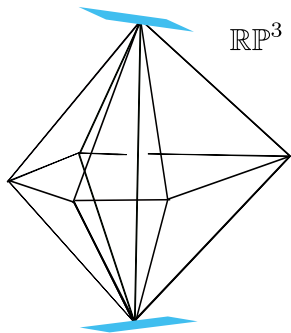


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$$G_2 G_3 G_4 G_5(T_1)$$

Projective gluing equations

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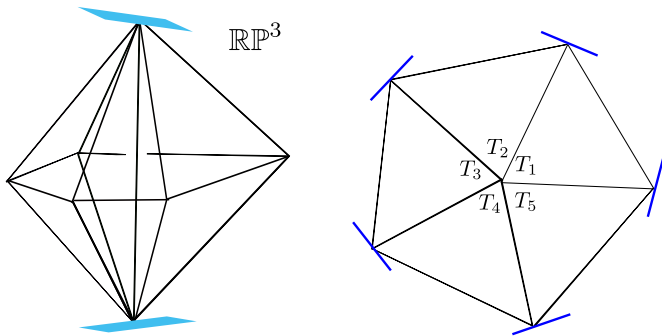


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In general $\prod_i G_i$ preserves a pair of incomplete flags

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G_i are **local** (depend on coordinates near face)

Equations can be read off from triangulation

Projective gluing equations

Given an orientable 3-manifold M with an ideal triangulation $\mathcal{T}$ we get a system of real equations

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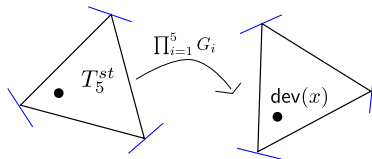
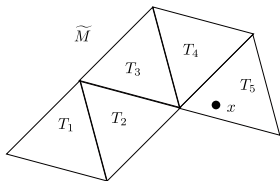
Each geometric solution give rise to a projective structure on M .

Projective gluing equations

Developing/holonomy

Each solution to the projective gluing equations gives rise to a **projective structure**:

- A developing map $\text{dev} : \tilde{M} \rightarrow \mathbb{RP}^3$

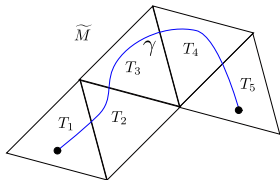


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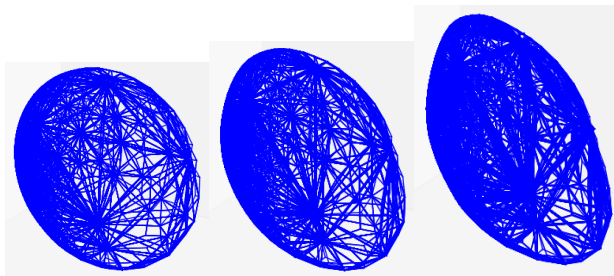
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- A holonomy representation $\rho : \pi_1(M) \rightarrow \text{PGL}_4$.



$$[\gamma] \mapsto \prod_{i=1}^5 G_i$$

Computations

Since the G_i are local, numerically calculating solutions and drawing pictures is possible



Projective gluing equations

Nice properties

- Get lots of interesting projective structures: hyperbolic, anti de-Sitter, convex projective
- Geometric properties of structures manifest as algebraic properties of solutions (**can tell if solution is hyp/ADS**).
- Numerically computing solutions can be automated (a la SnapPy)

Future directions

- Neumann-Zagier relationships (combinatorially defined redundancies in equations)
- Connections to quivers
- Geometric transitions
- Degenerations and tropicalizations

Thank you